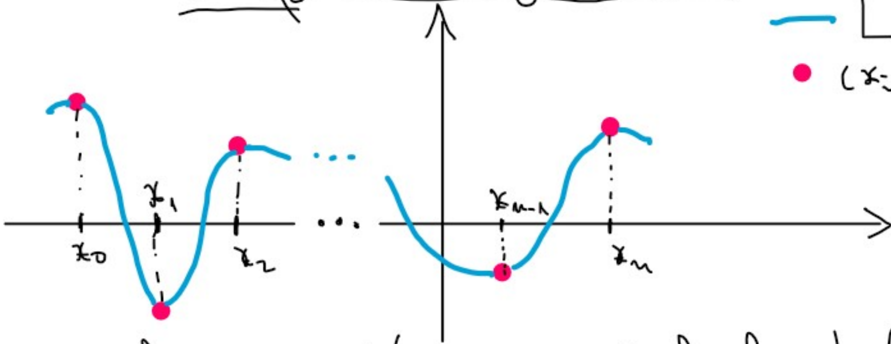
Lab 7-notite

Interpolare Lagrange cu formula Newton



— L

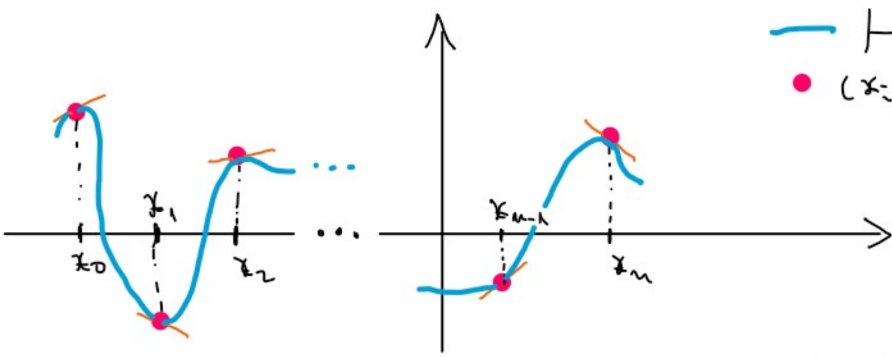
• (x_i, f_i)

$\overbrace{L(x_i) = f_i,\ i=\overline{0,n}}^{\text{conditie}}$

$\text{grad}\, L \le \text{nr. cond.} - 1$

Input: x, f, X Output: $L(X)$

Interpolare Hermite cu noduri duble



— H

• (x_i, f_i)

$H(x_i) = f_i$

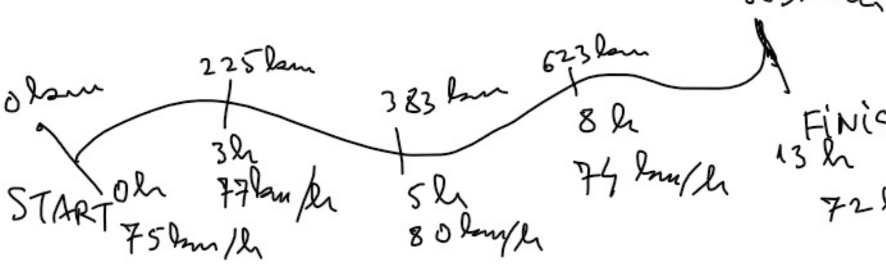
$H'(x_i) = df_i$ $i=\overline{0,n}$

$\text{grad}\, H \le \underbrace{\text{nr. cond.} - 1}_{2n+1}$

Input: x, f, df, X Output: $H(X), H'(X)$

x	Timpul	0	3	5	8	13
f	Distanţa	0	225	383	623	993
df	Viteza	75	77	80	74	72

viteza = derivata distanţei în fct. de timp



$$\underbrace{d'(t)}_{v(t)} = \lim_{\Delta t \to 0} \frac{d(t+\Delta t) - d(t)}{\Delta t}$$